A 3 years old girl presented to the casualty with fever, cough, and cold for 3 days, with no significant past history except for 2 attacks of respiratory infections during the last 2 months. When she was examined by you, she looks sick, febrile, and with widespread petechial rash all over her body.

1. List 4 signs on clinical examination to be checked in relation to her condition.
    - Vital signs
    - Sings of respiratory distress
    - Sings of meningeal irritation
    - Sings of bleeding
      Lymphadenopathy

Her blood investigations revealed an Hb 4.5 g/dl, WBC 1.8 X 103/mm3, platelets 25 X 103/mm3.

2. What is the most likely diagnosis?
   Leukemia
3. Mention 2 other differential diagnoses.
   Lymphoma
   ITP
4. Write one important investigation to be done to confirm the diagnosis.
   Bone marrow aspiration

Further investigations revealed that she has one type of malignancy and she was started on treatment.

5. Mention 4 complications of chemotherapy.
    - Bone marrow depression
    - Renal toxicity
    - Neuritis
    - Tumor lysis syndrome
    - Infertility.
    - Infection due to immune compromising
    - Hair loss
    - Nausea and vomiting

A five years old boy presented to the casualty with fever and convulsions for the last 3 days.

1. What 3 questions you want to ask the mother regarding his illness?
    - Type of convulsion
    - Duration of convulsion
    - Loss of consciousness
    - Sphincter status
    - Postictal state
    - Number of episodes in the last 24 hours
    - Is it the first time or not
    - Any family history

He was quite well before one week and started to feel unwell and not taking his meals as usual and became inactive. His parents

consulted a doctor in a health center and oral antibiotics were prescribed. There is no history of trauma.

2. Give 4 possible differential diagnoses for his condition.
    - Meningitis
    - Encephalitis
    - Cerebral malaria
    - Hypoglycemia
    - Epilepsy triggered by fever

Examination revealed temperature of 39.7º C, confused and irritable child, generalized body aches and pain including back and neck pain, and when neck flexed, he withdraws his legs to his abdomen.

3. Mention 2 other signs you want to examine in this patient.
    - Brudziniski sign
    - Neck stiffness

Lumbar puncture was done, CSF examination was as follows:
Cells: 100/mm3 mainly lymphocytes
Protein: 80mg/dl
Sugar: 40mg/dl-(random blood sugar: 100mg/dl)
Culture: negative

4. What is your final diagnosis?
    - Partially treated meningitis

5. Outline your plan of management.

#General ttt
Hydration
Paracetamol for fever .
Control of convulsions
#Specific ttt

Start empirical Abs and then according to culture and sensitivity give certain

6. How can you explain lymphocytosis and negative culture of CSF?
    - The prior treatment with antibiotics

Ali is 6 year old child, brought to the causality with a complaint of drowsiness and deep breathing. The mother gave a history of ill health and loss of weight for the last month. She also said that he has enuresis although he had been dry at night since the age of two years.

Q1. Write two other symptoms that you would ask for in the history

- Polydipsia, polyphagia and abdominal pain

On examination: he is dehydrated, has acidotic breathing with special breath smell.

Q2. What is the most likely diagnosis?

- DKA

Q3. Write three investigations that you would order

- Blood glucose
- ABG & Electrolytes
- Urine for sugar and acetones
- Hb A1C

Q4. Outline the management

- ABC
- Correction of dehydration
- Insert 2 large pore cannula
- Correction of blood sugar

- Correction of electrolyte disturbance
- Find and treat precipitating factor

He was controlled. But after 2 weeks he was brought to hospital comatose and convulsing.

Q5. How do you approach?

- Exclude hypoglycemia first
- Then you can give mannitol half rehydration until condition improve
- May need ICU admission

His blood glucose was 20.

Q6. What could have happened? What phenomenon?

- Honeymoon phenomena

Waleed is 7 years old. He started his second year elementary school 2 months ago. Last year he was the top of his class. The school master had phoned the family that Waleed has inability to walk. There was no trauma. Parents took him to Omdurman paediatric hospital. The doctor took history and examined him. There was symmetric weakness in the lower extremities.

Q1. Write 4 differential diagnoses of such weakness.

- GBS
- Trauma (excluded here)
- Transverse myelitis
- Tumor
- Granuloma

Waleed was admitted to hospital. Next day he developed urine retention and had to be catheterized. In 3 days time his condition deteriorated, as he

was unable to use his upper extremities. There was sensory loss on feet and hands.

2) Write 5 important investigations which should be done

- MRI spine
- CSF analysis
- EMG and NERVE conduction studies
- Serology
- Stool for poliomyelitis

The doctor was monitoring his vital signs every 4 hours including forced vital capacity and gave him intravenous immunoglobulin.

3) What were the vital signs which the doctor was monitoring?

- RR
- PR
- Temperature
- HR
- Oxygen saturation
- BP

4) Mention 3 clinical signs which should be observed frequently to recognize the need for intubation.

- RR
- Oxygen saturation
- Paradoxical breathing or abdominal breathing
- Chest expansion
- Signs of respiratory distress

Waleed CSF result came showing proteins of 500 mg/dl, sugar 75 mg/dl, cells were 3/HPF predominantly lymphocytes.

5) What is the specific diagnosis?

- GBS

6) Mention 2 complications which may occur

- Cranial nerve palsy e.g. Bulbar palsy
- Respiratory failure

7) What is the prognosis?

- Good prognosis
- 80% recover within months
- 20% with residual weakness
- Some will suffer relapse

A 7 years old boy from Ombada presented to Omdurman Emergency hospital for children with a history of fever for 3 days and dark urine for 2 days.

On examining him: he looks unwell, febrile, temperature: 38.5° C, respiratory rate: 24 breath/minute, pulse rate: 80/minute, BP 140/85

Other systems: no abnormality detected

Lower limbs examination revealed patchy skin lesion

Urine general revealed uncountable RBC, pus cells and granular casts

Abdominal ultrasound was normal

1) What is your diagnosis?
   - Acute glomerulonephritis
2) Mention 4 other investigations you will order.
   - CBC
   - Renal function test
   - ESR
   - ASO titer and DNAS titer
   - C3 complements

On the second day of his admission in the hospital he developed an attack of generalized tonic clonic convulsion and shortness of breathing.

3) Mention 2 possible causes for his convulsion.

- Hypertensive encephalopathy
- Uremic encephalitis

Shortness of breath may be due to?

- Fluid overload and metabolic acidosis

4) Outline the management of his condition.

- Admission
- Bed rest
- Renal chart
- Correction of any electrolytes, acid base imbalance
- Fluid restriction if severe edema present
- Antibiotics for infective causes
- Antihypertensive therapy
- Tx of complications e.g. pulmonary oedema

5) Mention 4 indications for renal biopsy in his condition.

- Acute renal failure
- Nephrotic syndrome
- Absence of evidence of streptococcal infection
- Normal complement
- Deteriorating hematuria, proteinuria and renal function
- Persistence of low C3 level for more than 2 month after onset

The parents of a 3 years old child from White Nile State came to see you being concerned about their child. She had several episodes of febrile illnesses sometimes accompanied with epistaxis for the last 2 months. The child was given several courses of treatment with no satisfactory response.

On clinical examination, she is small for her age; her temperature is 39° C, and has an appearance that is less ill than the height of temperature would suggest.

She has a splenomegaly palpable well below the umbilicus, moderately enlarged liver, and she is pale.

1. Mention 4 possible diagnoses for her condition.

Note;

Huge splenomegaly is defined as 12 cm below the coastal margin or splenomegaly below the umbilicus

DD for huge splenomegaly

- Tropical splenomegaly syndrome or hyperreactive splenomegaly
- Hereditary spherocytosis
- CML
- VL
- Myelofibrosis
- In pediatrics add storage diseases… glycogenosis and lipidosis

The following investigations were done:

Hb: 6.8 g/dl

TWBC: 2500/mm3

Platelet count: 30,000/mm3

ESR: 50 mm/hr

Peripheral blood picture showed no abnormal cells

2. What is the most likely diagnosis?

- VL

3. Mention 2 investigations to confirm your diagnosis.

- Serology; rk39 and DAT test
- LN aspiration

- Splenic aspiration

A full term male infant was born at home. This was his mother's first pregnancy. At 14 hours of age he was noticed that his skin and sclerae were yellow. He appeared well and was breast feeding. Serum bilirubin was 9 mg/dl at 20 hours and he was given intensive phototherapy. Six hours later his bilirubin rose to 14 mg/dl.

1) Mention 3 important causes for this baby jaundice

- ABO incompatibility
- RH incompatibility
- TORCH infection

Can Rh incompatibility occur in first pregnancy?

- Yes, due to previous sensitization either by incompatible blood transfusion or abortion

Remember: jaundice on the first day is always pathological

Give other causes

- G6PDH deficiency
- hereditery spherocytosis
- Crigler Najjar syndrome

What infections besides TORCH?

- Hepatitis
- Malaria… congenital

2) Write 4 investigations to be done for him

- Blood group for baby and mother
- CBC
- Peripheral blood picture
- Serological test for TORCH
- Then I can do coombs test, sepsis workup

How do you assess neonatal jaundice clinically?

- Gestational age
- General condition of baby wt and cardiorespiratory assessment
- Pallor
- Organomegaly
- Extension of jaundice according to involvement of body part it correlate with billrubin level
- Bruises or purpura
- Urine and stool color

Can you know crudely the level of jaundice clinically?

- Yes according to body part involved
- Head only or head and chest for example
- If head only bilirubin is estimated to be 5 mg/dl
- Head and chest 10 mg/dl
- Head chest and abdomen 12mg/dl
- Whole body 15mg/dl

3) What developmental function particularly needs to be checked following his severe jaundice?

- Primitive reflexes

What is the risk of jaundice in the neonatal period?

- Kernicterus

What is kernicterus?

- Bilirubin encephalopathy due to deposition of unconjugated bilirubin in basal ganglia

What is intensive phototherapy?

- Expose baby to certain wave length of blue light

What is *intensive*?

- Double or triple phototherapy

Is this baby a candidate for exchange transfusion?

And why?

- Yes, because in the first day the rate of rise of bilirubin is more important the level

A 15-month-old boy present because his parents are concerned that he does not speak recognizable words, has never made babbling sounds such as "baba" or "dada", does not follow verbal commands, and does not respond to his name. The child was a full-term baby without hospitalization or frequent illnesses. He sat without support at 6 months and began walking at 12 months. He is active in the examination room, but he does not respond to his name or to verbal commands of his mother. He is a well-developed, well-nourished normal child otherwise.

1. What area of development is deficient in this child?
- Hearing
2. What is the most likely diagnosis?
- Deafness
3. What is the next step?
- Hearing assessment

Sara is 3year old child from Kordofan state presented with a complaint of recurrent attacks of jaundice since the age of 7 months.

1) Write three differential diagnoses
- Sickle cell
- Thalassemia
- Hypothyroidism

Her elder sister had a history of swelling of both hands and feet.

On examination: she is pale, jaundiced and has a spleen that is 5cm below the costal margin.

2) What is the most likely diagnosis?
- SCA
3) How are you going to confirm this diagnosis?
- Peripheral blood picture for sickle cells
- Hemoglobin electrophoresis
4) Outline the long term management of her condition.
- Folic acid
- Hydroxyuria
- Prophylactic antibiotics
- Vaccination against encapsulated bacteria

Why do we give folic acid?

- To prevent aplastic crisis

What are the absolute indications for blood transfusion in SCA?

- CVA
- Priapism
- Acute chest syndrome

A 3years old boy presented to the hospital with severe stridor. He was quite normal a few days before the attack.

1) Mention 5 questions you would like to ask the mother concerning the stridor
- Onset
- Duration
- SOB
- Cough

- Fever
- History of choking
- History of trauma

On the next day of his admission he developed bull neck

2) Mention 4 differential diagnoses.

- Diphtheria
- Subcutaneous emphysema
- Tonsillitis
- Epiglottitis
- Soft tissue edema due to trauma

Other differential:

- Quinsy
- Retropharyngeal abscess
- Infectious mononucleosis
- angioneurotic edema

#Note: bull neck is due to lymphadenopathy and soft tissue swelling

3) Mention a single investigation to confirm each differential diagnosis.

- Diphtheria >>> throat swab and culture
- Epiglottitis >>> blood culture and neck X-ray
- Subcutaneous emphysema >>> chest x-ray

#Note: in case of epiglottitis, better to avoid laryngoscopy, even spatula, why?

- To avoid cardiac arrest

How?

- Due to vagal stimulation

4) Outline the emergency management of his condition.
- ABC

A pt with severe stridor what do you do to relieve the obstruction?

First

- Positioning
- Head tilt
- Chin lift

An 18 months old boy presented to the casualty suffering from watery diarrhoea and fever for 2 days.

Q1. Mention 2 questions you want to ask the mother about his illness

- Frequency of diarrhea
- Is it watery or bloody
- Ability to drink

On examination, he was found to be lethargic with sunken eyes, temperature 38°C, respiratory rate 30/min, pulse 100/min, skin pinch returns back very slowly.

Q2. Classify his degree of dehydration

- Severe dehydration

Q3. Mention 3 investigations to be done for this child

- CBC
- RBG
- Stool analysis
- Urea and electrolyte

Four hours later, he developed convulsions which lasted for 3 minutes. Investigations showed: blood sugar: 80 mg/dl, sodium: 120 mEq/L, potassium: 2.2mEq/L, urea: 30 mg/dl, blood film for malaria: negative, white blood count: 22,000, CSF analysis: not significant

Q4. Mention 3 possible causes for his convulsions.

- Febrile convulsion
- Electrolytes disturbances Hypokalemic and hyponatremia
- Hypoglycemia

What diarrheal disease is specifically associated with convulsions?

Shigellosis; this diarrhea causes

- High fever
- Smelly stools
- Convulsions

What CNS complication of dehydration?

- Convulsions
- Brain ischaemia
- CVA
- Cerebral edema
- Sinus thrombosis e.g. sagittal sinus

This pt is 10 kg

How do you manage?

- IV or interosseous rehydration
- Fluid type; Ringer lactate or normal saline
- Amount; 100 ml/kg = 1 L
- 30 ml/kg in the 1st half an hour
- 70 ml/kg in the next 2 hours and a half

- Zinc supplementation
- Vitamin A

A 7 years old child presented to the hospital with recurrent attacks of jaundice over the last two years. The jaundice episodes were getting deeper and more prolonged. He had frequent attacks of epistaxis.

On examination he was oriented in time, place, and active. There is a moderate degree of jaundice. Abdominal examination revealed an enlarged non-tender firm liver with irregular surface and blunt edge. The spleen is enlarged 6 cm. there is a positive shifting dullness. Other systems are normal.

Investigations:

- Urine general: bile ++
- BF for malaria: negative
- Haemoglobin: 12.9g/dl
- Liver function test
  - Total bilirubin: 8.6mg/dl___ direct bilirubin: 5.3mg/dl
  - Total protein: 5.4g/dl___ serum albumin: 2.1g/dl
  - AST: 68 IU
  - ALT: 84 IU
  - Prothrombin time: 19 seconds (control: 13 seconds)

The doctor told the parents that his liver function is decompensating and he needs to be admitted for further investigations and management.

1) What evidences did the doctor have to support his judgment of a decompensated liver?

Clinical;

- Jaundice
- Ascites

Laboratory;

- Low albumin
- Low AST and ALT
- Prolonged prothrombin time

Note; in chronic liver disease enzymes may be normal or low

2) What are the possible factors contributing to his ascites?

- Hypoalbuminemia
- Portal HTN

3) Mention 4 further investigations he will order.

- Abdominal USS/ CT
- Serum ceruloplasmin
- Viral screening
- Liver biopsy
- Sampling from Ascites

4) Give 2 possible diagnoses.

- Chronic hepatitis B or C
- Wilson disease
- Alpha 1 antitrypsin deficiency
- Hemochromatosis
- Autoimmune hepatitis

A 3-year-old boy has a 20-day history of fever that spikes twice daily. He was diagnosed with otitis media on the fifth day of fever and was prescribed amoxicillin, but the fever persisted. The fever is associated with a

faint rash on the trunk and proximal extremities and complaints of "body aches". A chest radiograph is normal, but a complete blood count (CBC) shows leukocytosis, thrombocytosis and anaemia. He developed an aversion to bearing weight and continues to have fever to (39.2 oC), but otherwise he has normal vital signs. His examination is remarkable for scattered lymphadenopathy, hepatosplenomegaly, and mild swelling of his interphalangeal joints and knees.

What is the most likely diagnosis?

- Juvenile idiopathic arthritis

What is the best diagnostic test for this disorder?

Note; There is no specific test for JRA, But once you suspect it, a few tests are helpful.

Nelson textbook;

DIAGNOSIS JIA is a clinical diagnosis without any diagnostic laboratory tests. The meticulous clinical exclusion of other diseases and many mimics is therefore essential. Laboratory studies, including tests for ANA and RF, are only supportive or prognostic, and their results may be normal in patients with JIA

What is the treatment for this condition?

Stepwise approach

- NSAID
- Aspirin
- Anti-malarial
- DMARD

Parents worry about their 4-year-old son's ability to walk. He first walked at 16 months, but he was clumsier than expected; they were reassured by

another paediatrician that he would "outgrow it". He remains clumsier than his peers, falls during simple tasks, and has developed a "waddling" gait. Within the last months he experienced increasing difficulty arising from a sitting position on the floor.

a. What is the most likely diagnosis?
- Duchene muscular dystrophy

b. What is the diagnostic test?
- Creatine phosphokinase CPK
- EMG
- Muscle biopsy

c. What is the mode of inheritance of this disease?
- X-linked recessive



What is the sign in the photo?

- Pseudo-hypertrophy of calf muscle

A 12-month-old boy has a 1-day history of intermittent abdominal pain and vomiting. His reports a 2-hours history of "lethargy". In the emergency department, he is quiet (almost lethargic) between periods of intense crying and remarkable distress. On abdominal examination, you find the right lower quadrant devoid of bowel and a right upper quadrant, sausage mass

a. What is your diagnosis?
- Intussusception
b. What is your next step?
- Barium enema
- X-ray to confirm the diagnosis
- Refer to surgery

What vaccine is associated with this condition?

- Rota

MCQs questions

A 15-year-old girl presented with ataxia, dysarthria and incoordination of upper limbs since the age of 10. She is now wheelchaired. O/E, she has scanning speech, horizontal nystagmus, ataxia of upper limbs, spastic paraparesis with absent knee and ankle reflexes, extensor plantar response, and bilateral pes cavus.

What is the diagnosis?

1. Ataxia telangiectasia
2. Autosomal dominant spinocerebellar ataxia
3. Brain stem glioma
4. Friedreichs ataxia
5. Multiple sclerosis

Answer; Friedreichs ataxia

A 16 year old female was admitted with shortness of breath. She is known to suffer from asthma. Which of the following feature would suggest that it is severe asthma

1. Blood pressure of 156/80 mmHg
2. Pulse of 100 beats per minute
3. Pulsus paradoxus
4. Respiratory rate of 22 per minute
5. Temperature of 39 C

Answer; pulsus paradoxus

A 2 years old boy present with a long history of poor development, offensive diarrhea and vague abdominal pains. O/E he is on 10th centile for height and 3rd centile for weight. Has a protuberant abdomen. What is the most likely diagnosis?

1. Celiac disease
2. Cow's milk protein energy
3. Crohn's disease
4. Cystic fibrosis
5. Ulcerative colitis

Answer; celiac disease

A 6 week old boy presented with projectile vomiting following feeds. He has been perfectly well until now. Examination revealed that he is constipated and dehydrated. The most likely diagnosis is

1. Duodenal atresia
2. Diaphragmatic hernia
3. Hirschsprung's disease

4. Hypertrophic pyloric stenosis
5. Staphylococcal food poisoning

Answer; Hypertrophic pyloric stenosis

Which of the following is a feature babies born to mothers with diabetes mellitus

1. Anemia
2. Hypercalcemia
3. Hyperglycemia
4. Hyperthyroidism
5. Shoulder dystocia

Answer; Shoulder dystocia

If a female carrier of a completely expressed X-linked condition marries an asymptomatic male. Which if the following is statements is statistically true of their offspring

Answer; Half females will be carriers

Which of the following skills would be expected of a 7 month old infant but not a 5 months old infant

1. Crawls
2. Smiles socially
3. Controls his bowel and bladder
4. Sits unsupported
5. Raises head while prone

Answer; sits unsupported

A 14 year old is brought to the clinic because of tenderness of the right breast. Examination reveals a small firm nodule that feels like a hypertrophied breast tissue beneath the areola. The most likely explanation for this finding is

1. Hermaphroditism
2. Pseudo-hermaphroditism
3. Physiological gyencomastia
4. Androgen secreting adrenal tumor
5. Testicular tumor

Answer; physiological gynecomastia

You are treating a 7 months old boy with dehydration due to diarrhea. The serum sodium is reported as 185 mEq/L, At this point you should plan to

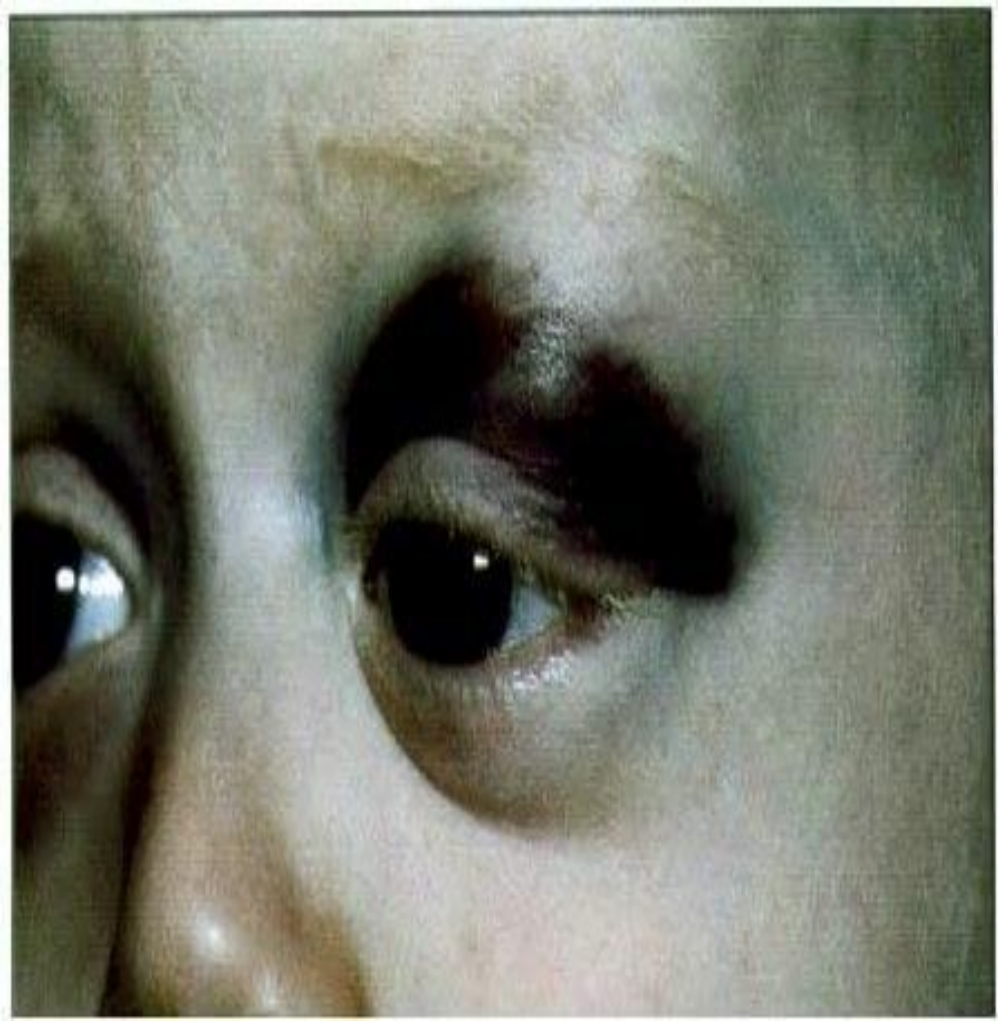
1. Use sodium free solution
2. Correct the hypernatremia slowly
3. Administer a hypertonic glucose solution
4. Administer mannitol
5. All of the above

Answer; correct hypernatremia slowly

A 2 years old boy presents with a large right flank mass, fever, weight loss, proptosis of the right eye, and ecchymosis of the right eye. What is the most likely diagnosis?

1. Wilm's tumor
2. Neuroblastoma
3. Hydronephrosis
4. Metastatic neuroblastoma
5. Disseminated tuberculosis

Answer; Metastatic neuroblastoma



Metastatic neuroblastoma

Parents bring their 5-year-old daughter to the clinic because she has developed breast and pubic hair over the past three months. Physical examination reveals a girl whose height and weight above the 97th percentile. Tanner stage II, breast and pubic hair development, oily skin, and facial acne.

a) What is the most likely diagnosis?
- Precocious puberty
b) How to evaluate this girl?

- Physical exam
- Family History
- Run blood tests to measure hormone levels

What hormone should be measured?

- GnRH
- LH
- FSH

What is Tanner staging?

Or SMR?

Stages of breast and pubic hair development

SMR: Sexual Maturity Rating

Or Tanner

In female: breast and pubic hair

In male: genitals and pubic hair

It is a clinical exam that should be done in any adolescent

Problems could be: early puberty or delayed puberty

Early puberty like our patient could be due to

- Most of the cases are idiopathic
- But there are predisposing factors
- It can also be due to estrogen secreting tumor

A 6-month-old child arrives for a well-child examination. His family recently moved to Khartoum State from Yemen. His medical and family histories are unremarkable except that his sole source of nutrition is goat's milk. He appears to be healthy on examination.

a. What hematological problem is most likely to develop?
- Folic acid deficiency and megaloblastic anemia

b. What non hematological concerns are considered in a child fed goat's milk?
- Lactose intolerance due to goat’s milk allergy leading to post-enteritis malabsorption

A mother says her-2-year-old daughter has had 1 to 2 weeks a history of perineal and peri-anal itching. She notes that the itching occurs mostly at night, but she denies fever, diarrhea, or emesis. The girl spends time in a “Mother’s Day out" program 3 days per week but otherwise is always with her mother. On examination, the perianal area is red and irritated; the anal sphincter tone is normal, and you find no evidence of penetrating trauma. The perineal area is similarly red and excoriated. Other than slight whitish vaginal discharge, the child diaper area is clean.

a. What is the most likely diagnosis?
- Entrobius vermicularis
- Pinworm infection

b. How can you confirm the diagnosis?
- Microscopy of perineal skin swab

c. What is the best management of this condition?
- Treat all contacts
- Drugs; Albendazole or mebendazole